

Sample Syllabus: Introduction to Data Science

Course Information

Subject: Introduction to Data Science (DS101)

Instructor: Dr. Jane Doe

Prerequisites: Basic Statistics and Python Programming

Topics Covered

1. Data Science Overview: Lifecycle, applications, and real-world use cases.
2. Data Collection & Cleaning: Data preprocessing, handling missing values, and data transformation.
3. Exploratory Data Analysis (EDA): Visualization, summary statistics, and pattern detection.
4. Statistical Methods: Probability distributions, hypothesis testing, and correlation analysis.
5. Machine Learning Basics: Regression, classification, clustering techniques.
6. Data Visualization Tools: Matplotlib, Seaborn, and dashboards.
7. Big Data & Tools: Hadoop, Spark, and data pipelines.
8. Advanced Topics: Deep learning basics, NLP, and model deployment.